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Research Article

Application and Optimization of Artificial Intelligence Technology in Architectural Design

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In the era of rapid development, the demand for architectural design has surged. In this regard, artificial intelligence technology is integrated into architectural design. The application and optimization of artificial intelligence technology in architectural design is studied by comparative analysis method. The research is as follows: (1) there are many shortcomings in architectural design, and the cost control disconnection has the greatest impact through comparison. Simply adding on-site inspections to architectural design can better complete architectural design. (2) For the application of artificial intelligence in architectural design, through the $F1$ performance index, it is concluded that neural network technology is the most useful for architectural design, and its accuracy is also the highest. After the application of neural network technology, all indicators of architectural design are there is a big improvement. (3) Comparing several optimization methods in terms of optimization, it is found that the optimization effect of focusing on design diversity through neural network is the best.

1. Introduction

Integer programming benefits from many innovations in models and methods studied for the application of artificial intelligence techniques in architectural design. Some promising directions for future elucidation of these innovations can be viewed in terms of a framework linking the perspectives of artificial intelligence and operations research. To demonstrate this, four key areas are investigated: (1) controlled randomization, (2) learning strategies, (3) induced decomposition, and (4) tabu search. Each of these has been shown to have features that seem usefully relevant to upcoming developments. A Guide to Soft Computing Artificial Intelligence Intelligent Systems Michael Negniewitzky almost all literature on artificial intelligence is expressed in computer science terms, filled with complex matrix algebra and differential equations [1]. Unlike many computer intelligence books, it turns out that most of the concepts behind intelligent systems are simple. Developed from student discourse with little knowledge of computing, these readers need no preconditions regarding knowledge of programming languages. Through a number of courses offered by the author, the methods used in the book are extensively examined. The field of computer intelligence has been introduced to include rule-based experience systems, fuzzy experience systems, framework-based experience systems, artificial neural networks, developmental systems, computing, hybrid intelligent systems, knowledge engineering, and data mining [2]. In artificial intelligence, the process of searching for a solution to a problem can be implemented without domain knowledge in many cases and with domain knowledge in other cases. The last process is often called heuristic search. In such an approach, matrix techniques are essential to reveal themselves. Their introduction can provide an easy and precise way to find solutions and explain how matrix theory emerges and engages productively in artificial intelligence, with viable applications in game theory [3]. In both collaborative efforts, architects rely heavily on well-designed coordination practices and artifacts. Based on an in-depth study of architectural work, these practices and artefacts are analyzed and shown to be multilaterally interrelated and form a complex of interrelated

practices and artefacts, which we call "sequencing" system. In the process, a conceptual framework for investigating and conceiving such practices is outlined [4]. Architectural design and urban planning projects often involve many architects and other professionals. Although design and review meetings typically involve large numbers of participants, design is still done by people using desktops and CAD applications between those meetings. A truly collaborative approach to architectural design and town planning is often limited to preliminary paper sketches. To overcome these limitations, the Arthur system (Augmented Reality (AR) Roundtable) was designed and implemented to assist architects in complex planning and design decisions. Although AR has been used in the field in the past, our approach is not intended to replace the use of CAD systems without seamless integration into a collaborative AR environment. This approach is complemented by an intuitive interactive mechanism that can be easily configured for different application scenarios [5]. The fields of architectural design and film theory are considered here in order to inform the design of virtual environments (VEs). It has been suggested that these domains could form the context for considering possible metaphors for designing VEs. First, the relationship between architecture and virtual reality technology is investigated through the nature of drawing and virtual environments as means of representing three-dimensional space. Then, by comparing the VE and the physical environment (PE) with our familiar everyday spatial experience, identify the differences between the VE and the physical environment (PE) to understand the inner nature of the VE. This step is crucial in helping us understand how to develop the architectural concept of the design space in the context of VE [6]. Architectural innovation is an emerging research field in the field of architectural theory in my country, which aims to study the theory and method of architectural design innovation. Extenics provide new theories, methods, and formal description tools for the field. The dissertation spans extenics, innovation, thinking science, and architecture to study the application of four extension thinking modes in architectural design innovation and then forms the theory and method of architectural design innovation based on extenics. There are diamond thinking method, inverted diamond thinking method, conjugate thinking method, and conduction thinking method. This paper discusses the relationship between the four extensional thinking modes and architectural design innovation and discusses the possibility of applying them to architectural design innovation [7]. Seven CBD systems were analyzed, experienced architects and design educators were interviewed, and an experiment was conducted to investigate how case engineers affect students' design performance. The results of this research show that although CBD has a good view of how architects acquire knowledge (planning), it is limited in important respects: it reduces building design to problem solving, is difficult to implement, and must go along with it. Objectives of Group K: fight prejudice. To extend these limitations, the design, implementation, and evaluation of Dynamo (Dynamic Architecture Online Memory) are covered [8]. This is one of the few reports that reflects the ongoing work of the Software Engineering Institute (SEISM) to understand the relationship between quality requirements and architectural design. The ultimate goal of this work is to provide designers with analysis-based guidance to make common design-quality features more reliable and easier to understand. To achieve this goal, four different problems must now be solved: (1) exact specification of quality attribute requirements; (2) determination of architectural decisions to be used to achieve desired quality attribute requirements; (3) relevant architectural decisions; (4) of relevant architecture in the design a method of incorporating decisions. Integrating solutions to these four issues into a design approach that is sensitive to business priorities is another challenge [9]. A framework for maintaining multidimensional design consistency is proposed. In a multidisciplinary design, each stakeholder creates their own part of the design. The part of the design for each stakeholder is called the stakeholder view. In order to express his opinion, the person has his own opinion. This perspective defines support for design concepts, symbols, and tools used by stakeholders. The framework proposed in this paper focuses on the design of heterogeneous architectures for distributed systems. A distributed system is a system whose components run on nodes of different physical systems. In such a system, the interaction of system components plays an important role. An example of a distributed system is a mobile communication network. In such a network, different parts of the system are located, for example, on the user's mobile phone, on the desktop computer of the network operator's employees, and on the mobile access point. Architectural design is a field of design that focuses on a higher level of abstraction in the design process [10]. In architecture, design begins with the development of an idea and its implementation in a concrete spatial form. Architects define design problems by creating and testing alternatives to achieve the desired spatial form. A comprehensive knowledge of architecture helps an architect. This knowledge is a combination of practice and theory, that is, mystery and doubt, intuition and science, and experience and research. The architect has to do this in two ways, creatively integrating all the components. Scientists and research methods aim to study the contribution of spatial syntax to the design process. The spatial syntax is based on the theory of spatial configuration and attempts to decipher the morphology of space and its impact on human activity. With the development of new techniques for describing and analyzing spaces, spatial syntax provides architects with a tool to explore their design ideas and understand a design's potential impact [11]. Each floor plan can be viewed as a reflection of the user's goals and activities as interpreted by the architect. Comparing different building plans of similar institutions provides a good understanding of how goals and values are reflected in spatial solutions and allows for the development of spatial functional typology of design solutions. Postuse evaluation focuses on the main arguments and user experience of different design solutions, providing insight into relevant decision points, usability, perceived (weak) advantages, and complexity of spatial and social systems, showing how the integration of floor plan benchmarks and end-of-use evaluations can help

provide more robust solutions for programming and architectural design [12]. A number of current research and design projects are combining Computer Aided Architectural Design (CAAD) and Computer Aided Architectural Manufacturing (CAAM) programming. The result of our work is a proof-of-concept construction project that demonstrates these technologically based architectural concepts. The conclusions of these projects work on two levels: a set of conclusions about the implementation of new technologies and processes by architects, and secondly, how these technologies can be presented to the different experiences of architects and construction professionals and optimally integrated in the work. It uses its own production methods CNC (Computer Numerical Control) and the skills of traditional architects [13]. Symmetry and regularity abound in architectural design, often for economic, industrial, functional, or aesthetic reasons. It demonstrates how recent work in symmetry and texture detection can be used to analyze digitized real-world artifacts using architectural sketches and 3D scanning techniques. It enables reverse engineering of program models, allowing efficient exploration of the underlying design space and synthesis of new models by tuning the resulting structure and symmetry parameters. The effectiveness of this method is shown in several implementation examples [14]. Digital fabrication is impacting the architectural design process as it plays an increasingly important role in the production of architectural models. Many design professionals, professors, and students have experienced the benefits and challenges of using digital fabrication in the design process, but many others in the industry are unaware of the opportunities and drawbacks of these technologies [15].

2. Application of Artificial Intelligence in Architectural Design

2.1. Architectural Design. Architectural design means that before constructing a building in accordance with the task of construction work, the author of the design makes a general assumption about the problems that arise during the construction process, implements procedures, finds ways to solve these problems, plans, and submits drawings and documents. Production is the common basis for the supply of materials, organization of construction work, and various forms of cooperation in the construction industry. The advantage is that the entire project runs smoothly and consistently according to the plan within the set investment. And get ready-made products that fully meet the various needs and applications expected by consumers and companies. The so-called architectural design is the use of "Virtual Reality" technology in urban planning, architectural design, and other fields. In recent years, urban roaming animation with features that cannot be achieved by traditional methods, such as human-computer interaction, realistic design, and large-scale 3D terrain simulation, has been widely used at home and abroad. With City Walk animation, a complete assessment of future buildings or urban areas can be performed dynamically and interactively in a 3D virtual environment: scene details can be viewed from all angles, distances, and angles. You can freely switch between walking, driving, flying, and other sports modes to surf. In addition, real-time changes can be made while roaming to compare different design options and their environmental impact. It can provide users with strong and realistic sensory effects and a vivid experience. There are many types of architectural models in Figure 1.

2.2. Introduction to Artificial Intelligence. The computing field is more concerned with artificial intelligence, applications in robotics, economic decision-making and politics, operating systems, and simulation systems. Professor Nelson's motto is: "Cognition is the scientific method - the science of presenting, acquiring and applying knowledge. How can computers perform intelligent functions that only humans could perform before?" The basic idea and principle of artificial intelligence is that artificial intelligence is a study of knowledge patterns and is the study of artificial systems with certain knowledge. Computer technology is the predefinition of human intelligence and learning certain computer capabilities, that is, how to use computer science and hardware to compare some basic principles, behaviors, and methods of human cognition. Artificial intelligence has been an industry since the 1970s and is one of the top three technologies in the world. Knowledge covers areas such as computer science, psychology, philosophy, and languages, almost every area of the natural and social world outside of information technology. The link between AI and better thinking is the link between practice and theory. False intelligence is the science of the mind. At its level is its application. From a rational perspective, critical knowledge is not limited to simple logical thinking; AI supports visual and heuristic thinking. The main application areas of artificial intelligence today are as follows in Figures 2 and 3.

Figure 2 describes the main distribution areas of artificial intelligence technology, mainly illustrating the application layer, technical layer, and basic layer of artificial intelligence technology. Applied technologies include computer vision, natural language processing, and speech recognition, and the applied algorithms include machine learning, deep learning, and reinforcement learning.

2.3. Application of Artificial Intelligence in Architectural Design. There are many aspects to the application of artificial intelligence in architectural design, the data in architectural design will be understood through different algorithms, and understanding the terms of this data will help you decide what information to submit. Compare automation with traditional automated scoring methods to assess whether data on input pages meets human-specified criteria and display results on output pages. Using artificial intelligence algorithms to collect information from data requires strong computer training and self-learning capabilities, as well as the ability to create mathematical models and analyze various scenarios. Compared with the traditional architectural design industry, artificial intelligence-based architectural design provides the world with a more efficient working environment. Artificial intelligence algorithms determine different scenarios. After the architect has

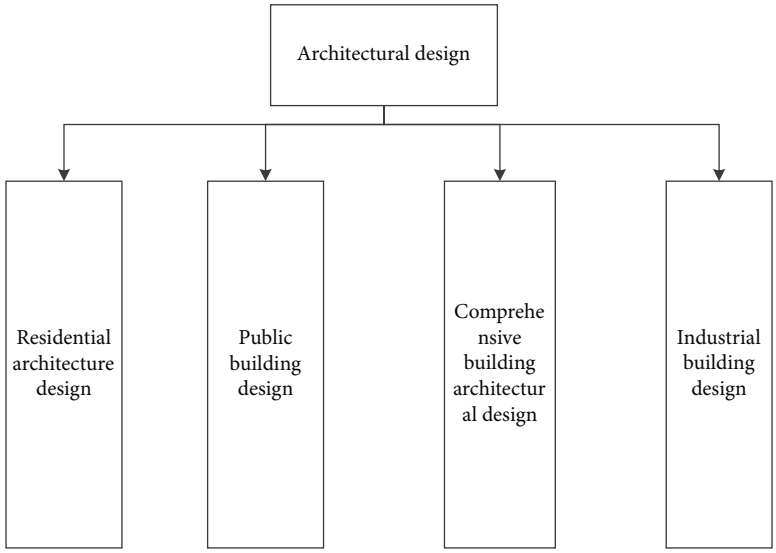

Figure 1: Classification of architectural design types.

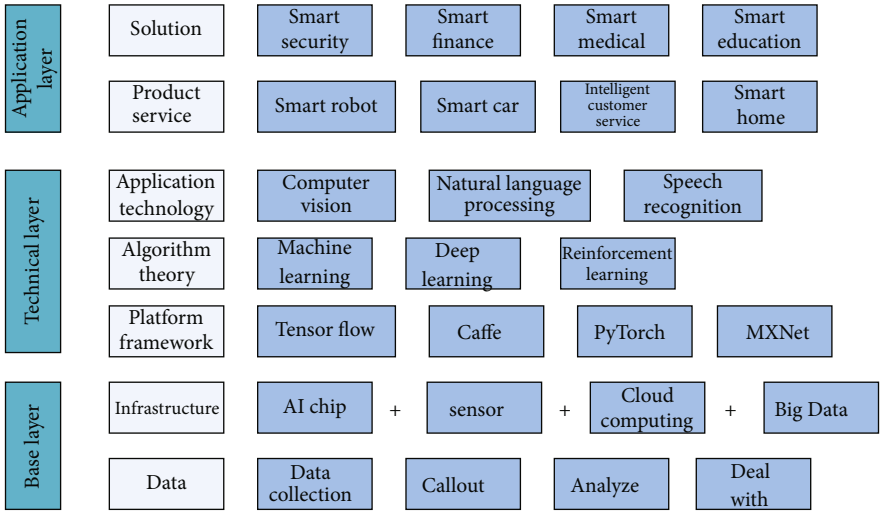

Figure 2: Main applications of artificial intelligence.

determined the financial and technical parameters such as location and site requirements, the software will quickly calculate the recommended building combination, and effectively adjusting various parameters can greatly improve the project efficiency, WYSIWYG on-site demonstration. For example, you can also check for errors in the project design. The AI design platform provides progress analysis results and generates progress reports for each process.

2.4. Neural Network Structure. Neural networks are complex networked systems composed of extensive interconnections of large numbers of simple processing units called neurons. Nonlinear Dynamic Learning Systems: it has the characteristics of large-scale parallelism, distributed storage and processing, self-organization, self-adaptation, and self-learning, especially to solve the incorrect and chaotic elements and conditions of information processing problems, and is especially suitable for execution to obtain a large amount of information at one time.

3. Architectural Design Algorithms

The application of artificial intelligence technology in architectural design is mainly used in the optimization of the surrounding environment, the use of the site, the optimization of the architectural form, and the choice of architectural style through the neural network model and the genetic algorithm.

3.1. Structural Relationship of BP Neural Network. We have discussed the general structure of artificial neural networks; we can create neural networks of different structures (the structure here refers to the connection of two neurons), i.e., neural networks with multiple hidden layers. In this configuration, it is easy to calculate the initial value of the neural

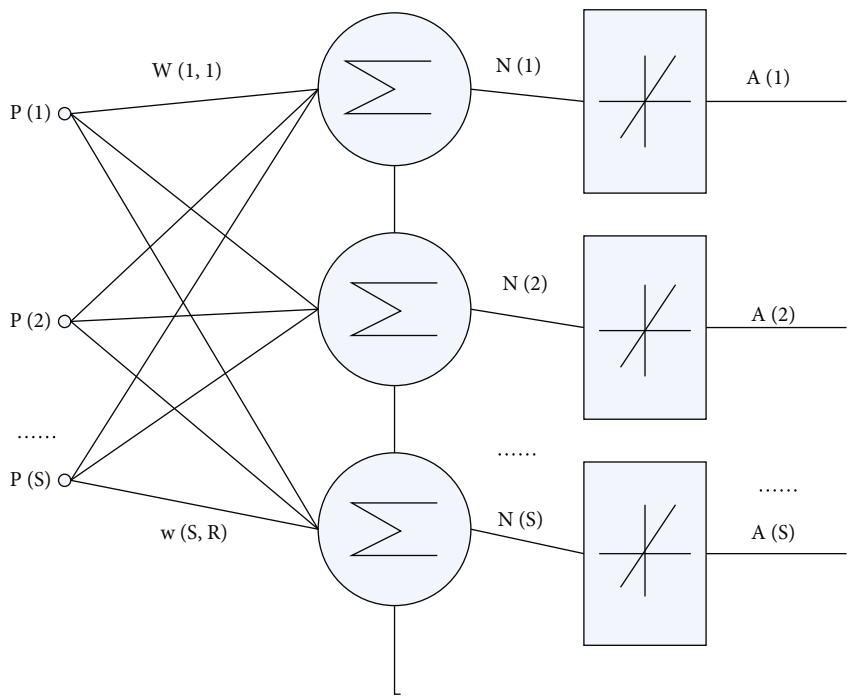

Figure 3: Linear neural network architecture.

network; we can follow the formula deduced earlier and propagate forward step by step, calculating each activation value of the L2 layer cell by cell, and then, the activation value of the first L3 layer to the end of an interlayer. This link graph has no loops or closed loops, so this type of neural network is called a feedforward network.

In addition, the neural network may have more than one output device. For example, the neural network structure in Figure 4 has two hidden layers: (layers L2 and L3), and the output layer L4 contains two output devices.

(1) Determination of input neurons and output neurons

The neural network starts to expand as input parameters are fed to the neurons of the input layer, and each input parameter is assigned a certain weight as it enters the neurons of the hidden layer. The input and output methods are as follows (1) and (2). The expected output vector corresponding to the input pattern is:

$$\begin{aligned} X &= (X_1, X_2, \cdots, X_n), \\ Y &= (Y_1, Y_2, \cdots, Y_s). \end{aligned} \tag{1}$$

In this paper, for the optimization and single reference information in architectural design, the input neurons are set to 29, namely, latitude (north latitude), longitude (east longitude), altitude, orientation, function type, building classification, structural form, building area, building height, proportion of heating and cooling rooms, roof insulation material, thermal conductivity of roof insulation layer, thickness of roof insulation layer, external wall insulation material, thermal conductivity of external wall insulation layer, thickness of external wall insulation layer, main wall material, interlayer floor slab thermal insulation material, thermal conductivity of interstory floor thermal insulation layer, comprehensive window-to-wall ratio, exterior window frame material (K value), exterior window glass material (K value), exterior window heat transfer coefficient, visible light transmittance, and window air tightness grade. The number of output neurons is set to 1, and the output value is the optimization degree of architectural design. There is a one-to-one correspondence between input data and output data, where n is the number of units in the input layer and s is the number of units in the output layer.

(2) Determination of activation function δ

The activation function of the output node of the BP neural network can use several different functions, the first is a linear function, that is,

$$f(x) = k * x + c. \tag{2}$$

The second type is the ramp function, which is

$$f(x) = \begin{cases} T, & x > c, \\ k * x, & |x| \le c, \\ -T, & x < -c. \end{cases} \tag{3}$$

The third is the threshold function, which is

$$f(x) = \begin{cases} 1, & x > c, \\ 0, & x \le -c. \end{cases} \tag{4}$$

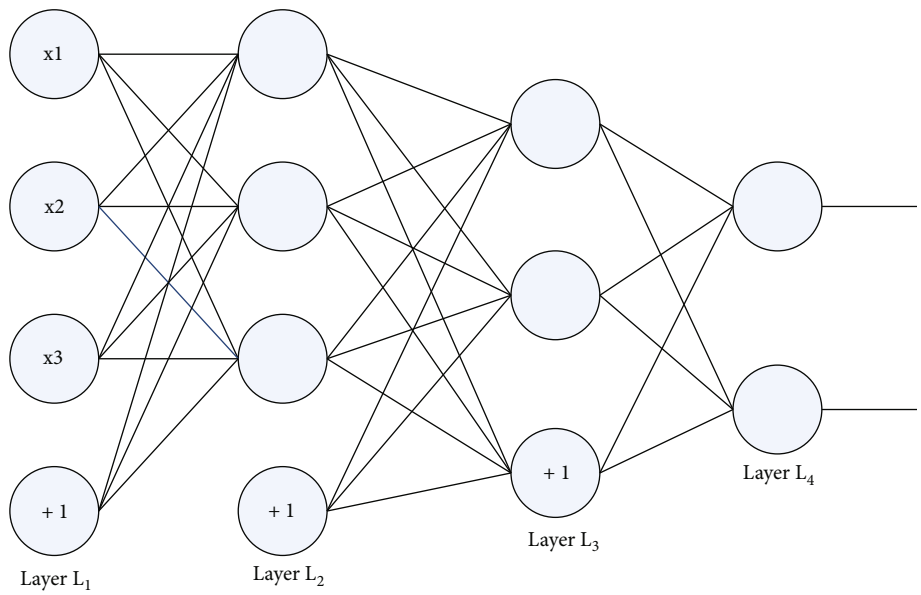

Figure 4: Neural network connection diagram.

The fourth type is the sigmoid function (sigmoid), which has the following two forms:

$$f(x) = \frac{1}{1+e^{-ax}} (0 < f(x) < 1),$$
$$f'(x) = \frac{\alpha e^{-ax}}{(1+e^{-ax})^2} = \alpha f(x)[1-f(x)]. \quad (5)$$

The last class is called bipolar sigmoid function:

$$f(x) = \frac{2}{1+e^{-ax}} - 1(-1 < f(x) < 1),$$
$$f'(x) = \frac{2\alpha e^{-\alpha x}}{(1+e^{-\alpha x})^2} = \frac{\alpha f(x)\left[1-f(x)^2\right]}{2}. \quad (6)$$

In the training and simulation process of the neural network model, the correction function formula is as follows:

$$f(x) = \frac{1}{1+e^{x/x_0}}. \quad (7)$$

Studies have shown that a single hidden layer BP neural network can continuously approximate any continuous function, while a three-layer BP neural network maps an n -dimensional neural network can. This paper examines the relationship with architectural design, so this paper fixes the number of layers of BP neural network to three layers and then fixes the hidden layer to one layer. In this single hidden layer, the number of neurons is usually determined by the number of input and output neurons. According to the training results, to determine the hidden layer, it contains the number of neurons in the layer. The empirical formula for the number of nodes in the hidden layer of the neural prediction model in this study is:

$$h = \sqrt{m+n} + a. \quad (8)$$

h is the number of hidden layer nodes, m and n are the number of input and output layer nodes, and a is a setting constant between 1 and 10.

3.2. Data Processing. When we learn neural networks, we also need to normalize the sample data, i.e., H. network input data and corresponding expected values, time normalizes them to a predetermined interval. After the network has been trained and trained, it must be prenormalized. Normalized predicted values are converted to actual values. In the architectural design, the collected data must be normalized to [0, 1]. Normalized linear transformation algorithm formula:

$$y = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (9)$$

In the formula: x represents the input variable x; $x_{\min}$ represents the minimum value of the input variable x; $x_{\max}$ represents the maximum value of the input variable x; y represents the normalized output variable; Equation (9) normalizes the data collected in the architectural design. For processing, it needs to be normalized to between 0 and 1 to facilitate subsequent calculations.

3.3. Application Performance Evaluation. To evaluate the architecture design optimization, it is necessary to introduce new variables relative error a, average relative error, and root mean square error RMSE and use the above three error metrics to perform error analysis and performance evaluation for architecture design optimization. The formula is:

$$a = \frac{\text{Design energy consumption} - \text{Forecast energy consumption}}{\text{Design energy consumption}} \times 100\%,$$

$$\text{Mean} = \sum_{i=1}^{N} \frac{a}{N},$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} a^2}{N}}. \quad (10)$$

3.4. Optimization Problems. For optimization problems, the situation is often very complex, and there are many types of objective functions and constraints. Before optimization, it is necessary to establish a mathematical programming model for the problem to be optimized. For a solving function minimization problem, the mathematical programming model is as follows:

$$\begin{cases} \min\ f(x), \\ \text{s.t.} X \in R\, R \subseteq U. \end{cases} \quad (11)$$

We need to consider many factors of the beam during construction. We set the beam section width as b_b and the beam section height as h_b. There are requirements for the aspect ratio of the beam, which should generally meet:

$$\frac{1}{3} \le \frac{b_b}{h_b} \le \frac{1}{2}. \quad (12)$$

For beams, the shear-compression ratio has a great relationship with the bearing capacity of the oblique section of the beam. In order to prevent the oblique pressure failure, it is necessary to ensure that the section size of the member cannot be too small. The minimum section size constraint of the flexural member can be expressed as the following formula:

When $(h_w/b_b) \le 4$, the constraints need to be satisfied

$$V_b \le 0.25\beta_b f_{cb} b_b h_{0b}. \quad (13)$$

When $(h_w/b_b) \ge 6$, the constraint needs to be satisfied

$$V_b \le 0.2\beta_b f_{cb} b_b h_{0b}. \quad (14)$$

When $4 < (h_w/b_b) < 6$, the constraints need to be satisfied

$$V_b \le 0.025\left(14 - \frac{h_w}{b}\right)\beta_b f_{cb} b_b h_{0b}. \quad (15)$$

For the constraints of beams, the building design can be optimized as much as possible.

4. Research on Application Optimization of Artificial Intelligence Technology in Architectural Design

4.1. Current Problems in Architectural Design. Architectural design refers to the design carried out to achieve the design purpose of a specific building. During the design process, architects often deal comprehensively and complexly with the role, function, and visual impact of a building in its surroundings. Building solutions can be divided into three categories according to design objects: civil building solutions, industrial building solutions, and agricultural building solutions. It is shown in Table 1.

Table 1 is the classification of the current architectural design. At present, the architectural design is mainly divided into two categories according to the design object and the design content. The design content mainly includes four categories: architectural design, structural design, physical environment design, and equipment design. Architectural design is represented by main body design and exterior wall design; structural design is represented by brick-concrete structure design and steel structure design; physical environment design is represented by acoustic design and thermal design; equipment design is represented by water supply and drainage design, electrical design, etc. The classification of architectural design can better help artificial intelligence technology to optimize.

Through the classification of architectural design, we can see that the design is divided into many types. At present, there are still many problems in architectural design, which need to be improved. A comparative analysis is made on the degree of impact of several serious problems and the frequency of occurrence of problems. The result is shown below:

From Figure 5, we can see that in the comparison of the shortcomings of the main four architectural designs, the greatest impact is the disconnection from cost control. We use segmented management to manage the investment, estimation, cost estimation, and budget of construction projects. And accounts are done on a project-by-project basis for each part of the cycle and are prepared in stages. The design department is generally responsible for the preliminary design and budget of the building and cannot estimate the cost of coordinating and modifying the building. His degree of influence has reached 91%. Poor cost control will lead to an increase in the cost of architectural design, and some even affect the effect of design. The degree of influence of lack of talents is relatively low at 67%, and the degree of influence of lack of innovation is 78%, and the degree of influence of insufficient technical and economic integration is 74%. In terms of frequency of occurrence, the number of occurrences of cost control disconnection is 7 times per day, which is the highest among these items. For the shortcomings of architectural design, there are mainly the following targeted solutions, recruit a large number of talents, actively carry out innovative design, and strictly control the cost in the design, and the economy and technology cannot be closely integrated. This situation can be effectively avoided. The architectural design satisfaction survey after several optimization methods are used is shown in the figure below.

Figure 6 introduces the satisfaction survey after optimization of the optimization methods for different methods of architectural design. The optimization of architectural design can make it easier to control when using artificial intelligence technology to optimize because simple architectural design is affected. For the four methods of optimization, the optimization degree of recruiting talents is the

Table 1: Classification of architectural design.

Classification by	Classification name	Concept definition	Main content
Design object	Public building	Buildings for public activities	Offices, hotels, hospitals, etc.
	Residential building	Buildings used for daily living	Residential, apartment, villa, etc.
	Industrial building	Buildings engaged in various production activities	Factory, warehouse, etc.
	Farm building	Buildings for agricultural and animal husbandry production and processing	Greenhouses, seed banks, etc.
Design content	Architectural design	Designs for the exterior and interior of buildings	Main body design, exterior wall design, etc.
	Structural design	Design of building materials, and stress systems	Steel structure design, brick-concrete structure design, etc.
	Physical environment design	Design for the physical properties of the building	Acoustic design, thermal design, etc.
	Equipment design	Planning and design of equipment used in buildings	Water supply and drainage design, electrical design, etc.

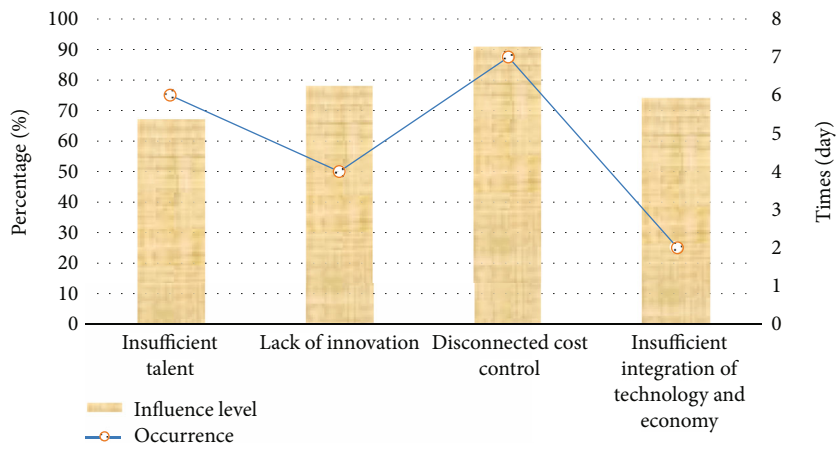

Figure 5: Comparison of architectural design shortcomings.

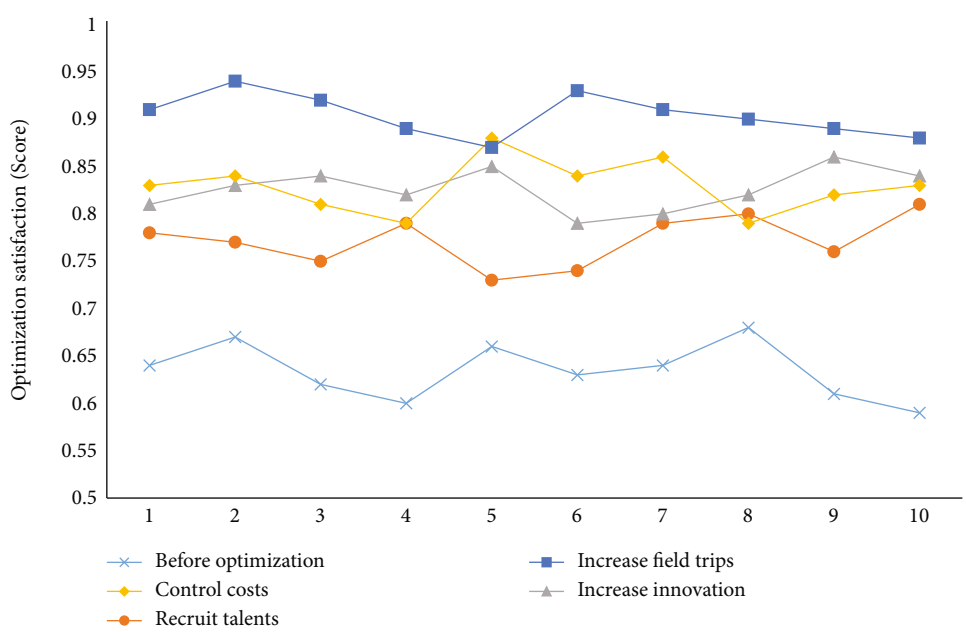

Figure 6: Satisfaction comparison after optimization by different methods.

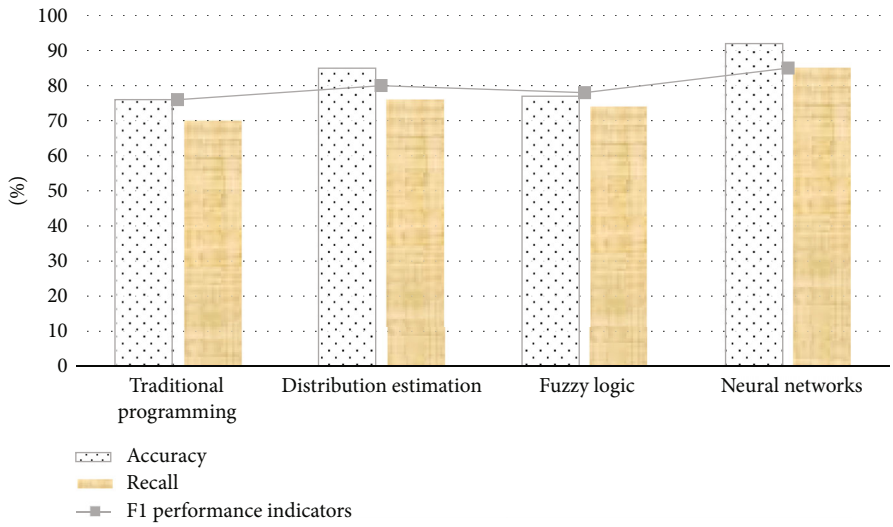

Figure 7: Performance comparison of artificial intelligence application technologies.

lowest, followed by increasing innovation, the optimization satisfaction of cost control is higher, and the average of recruiting talents is 0.8; the best degree of optimization is to increase on-site inspection.

For the architectural design satisfaction scores before and after optimization with different methods as shown in the figure, it can be seen that the satisfaction before optimization is the lowest, and after optimization with different methods, there is a great improvement, among which the optimization scheme of on-site inspection is added. The degree of satisfaction of the architectural design is the largest, with an average of 0.9. The satisfaction of the other methods for the architectural design is also around 0.8. It can be seen that the optimization effect is very obvious.

4.2. Application of Artificial Intelligence in Architectural Design. Introduction to Artificial Intelligence Architecture Software Artificial intelligence differs from simple automation. The biggest difference is that AI uses deep learning to extract common functions from large amounts of data and control their similarity and association rules. If a person to some extent learns the rules of this data, they can later correctly evaluate the entered information. Unlike traditional automated evaluation methods, automation aims to evaluate whether data on the input side meets manually defined conditions and present the results on the output side. The AI algorithm evaluates different scenarios. After the architect has entered key economic and technical data, such as the ratio of plot to floor area, the software provides recommended parameters for the combination of building types in the shortest possible time by effectively comparing calculations and various aspects to increase efficiency. Artificial intelligence mainly includes computer technology, control technology, psychology, and language technology. It is a systematic discipline based on several disciplines. At the same time, a large number of applied mathematical models and theories are needed to drive the rapid development of artificial intelligence technology. Currently, research on the application of artificial intelligence technology mainly includes the following three methods for realizing artificial intelligence technology: fuzzy logic, neural networks, distribution estimation, and traditional programming. For these methods that we mainly use, we also need to compare experimental data to observe their architectural design recall rate, design accuracy, and $F1$ performance indicators to monitor. We did experiments. Comparative experimental data are shown in the figure.

Figure 7 is a comparison of the performance of the introduced artificial intelligence application technology. The artificial intelligence technology in architectural design mainly includes four methods: traditional programming, distribution estimation, fuzzy logic, and neural network. Appropriate methods are applied and optimized. Figure 5 shows that the accuracy, recall, and $F1$ performance indicators are the highest in the neural network, indicating that its performance is the best. Neural networks should be chosen for architectural design.

From Figure 7, we can see that the best application of artificial intelligence technology in architectural design is neural network technology. The accuracy and recall rate of neural network technology in architectural design applications are both the highest, and the $F1$ index of performance comparison is also the neural network is the highest, so neural network technology should be used in architectural design. Whether the efficiency of architectural design is improved through artificial intelligence technology is also a key application factor. We have compared the architectural design engineering quantity index, project life, and cost per unit area under the action of artificial intelligence technology to observe its effect.

From Figure 8, we can see that the architectural design using artificial intelligence technology has reduced the amount of work on the same project, the cost per unit area has decreased, and the life after the completion of the project has also increased by 5 years. It can be seen that the artificial intelligence technology has a very important influence on architectural design. The main application areas of artificial intelligence in architectural design are shown in the figure below.

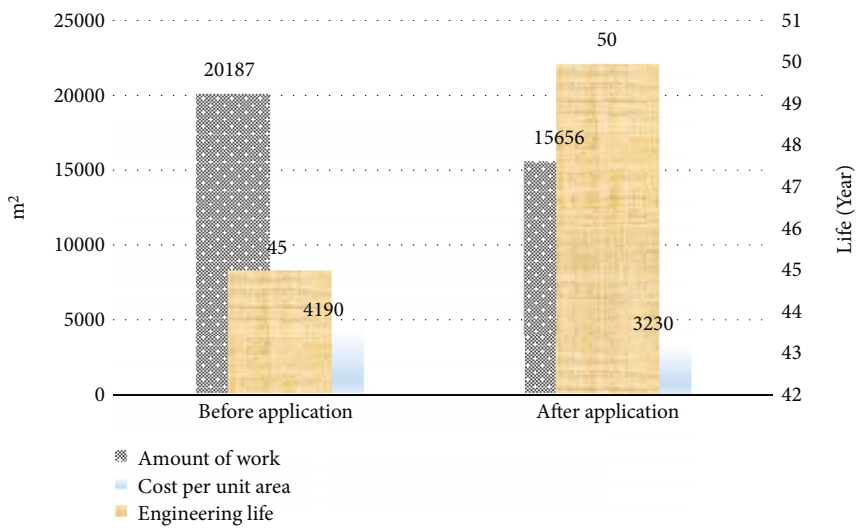

Figure 8: Comparison before and after the application of artificial intelligence technology.

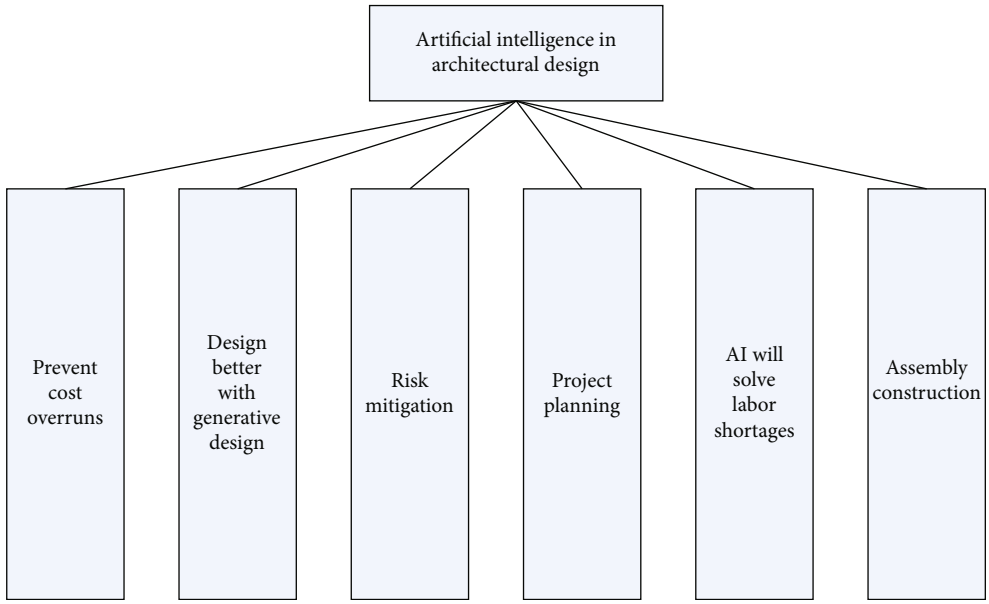

Figure 9: Application areas of artificial intelligence in architectural design.

Table 2: Issues with artificial intelligence technology in architectural design.

Aspect	Defect
System aspects	Low level of intelligence
Coordination	Insufficient coordination between intelligent technology and architectural design
Architectural aspects	Insufficient attention to building functional diversity
Environmental aspects	Serious waste of resources

From Figure 9, we can see that artificial intelligence technology is applied in many places in application, and most of them use neural network technology, which is very helpful to architectural design.

4.3. Optimization of Artificial Intelligence Technology in Architectural Design. The artificial intelligence technology in architectural design started late, but it has developed rapidly, with a certain range of intelligent building control systems, especially office automation systems, and intelligent communication control systems, such as automation systems and building automation systems. With the continuous development of intelligent buildings, higher and more comprehensive requirements are also put forward for the communication means, system automation, service requirements, and quality control functions of various control

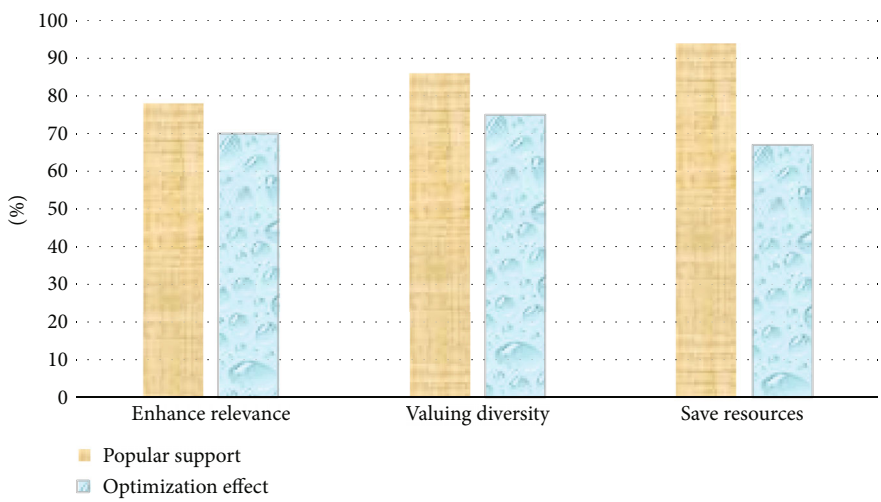

Figure 10: Comparison of optimization methods.

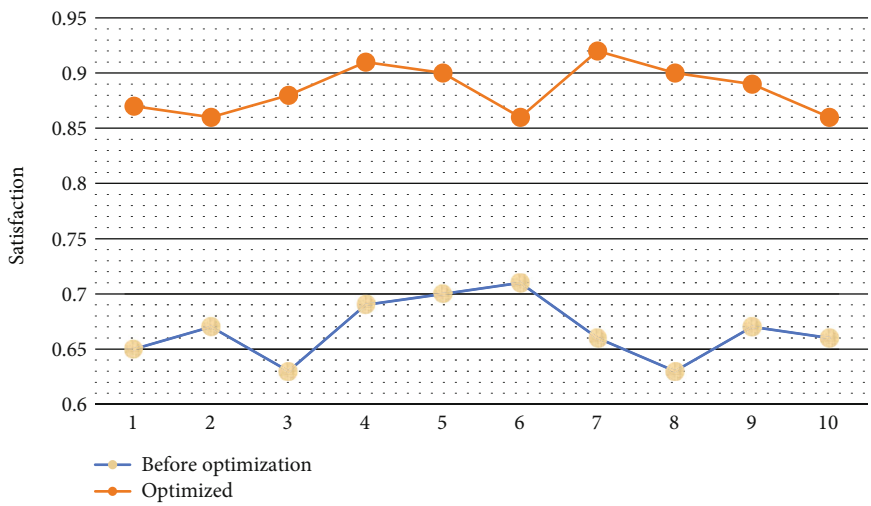

Figure 11: Comparison of satisfaction before and after optimization.

systems. This has also created a bottleneck in the current development of intelligent buildings. Although the development of smart buildings is advancing rapidly, there are still some problems that need to be optimized in the development of smart buildings today. The main problems at present are as follows in Table 2.

These problems need to be optimized, and the use of neural network technology in artificial intelligence technology can effectively solve the problem. The optimization suggestions include closely combining neural network technology with architectural design and integrating it into the design; paying more attention to the diversity of building functions; integrating energy; and using artificial intelligence technology to define classification and enhance recycling. For the optimized method, the satisfaction score of the optimized architectural design is shown in the figure below.

Figure 10 in the text is a comparison of the effects of artificial intelligence technology optimization methods in architectural design. For the comparison of the public support rates of the three optimization methods, the public support rate for saving resources is the highest, reaching 94%. Emphasis on diversity and enhanced correlation and sexual support was slightly lower at 86% and 78%, respectively. From the perspective of optimization effect, the optimization effect of emphasizing diversity is the best, which can better enhance the diversity of architectural design and make the public more satisfied.

From Figures 10 and 11, we can see that the best optimization method is to pay attention to the diversity of architectural design. Next, we compare the satisfaction of this method before and after optimization, as shown in the following figure.

From this figure, it can be seen that the average satisfaction rate after optimization is increased by 0.2, which shows that the optimization effect of this method is obvious, and it can play a good optimization effect.

5. Conclusion

This paper gives a comprehensive introduction to the application and optimization of artificial intelligence technology

in architectural design. Architectural design is a very important engineering work related to the lives of the public. The application of artificial intelligence technology in many aspects of architectural design has played a role in architectural design, convenience, and accurate effect; for its existing problems, this paper also finds out the optimization scheme and compares it and finds that the optimization effect is obvious. In the future, attention should be paid to environmental protection issues in architectural design, and while completing work efficiently, it will also reduce damage to the environment.

Data Availability

The experimental data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declared that there are no conflicts of interest regarding this work.

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